

基于神经网络的二类水体大气校正

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摘要: 以 MERIS 高光谱影像为数据源, 根据现有大气传输模型和大气校正方法, 探索了适合于内陆湖泊二类水体的高光谱遥感影像大气校正方法。在 6S 辐射传输模型的基础上, 构建了基于神经网络的二类水体大气校正算法。通过构建输入卫星辐亮度直接提取离水反射率的模型, 无需同步气溶胶参数, 即可实现大气校正。对 2010 年 8 月 9 日的 MERIS 影像进行大气校正, 并将校正后的遥感反射率与准同步实测离水反射率进行对比分析, 结果表明, 大气校正过程有效去除了大气效应的影响, 经过大气校正的 13 个波段的平均相对误差分布在 10%~40%, 得到了与实测值相近的水体遥感反射率。

关键词: 水色遥感; MERIS; 大气校正; 神经网络

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水色遥感是指利用可见光、近红外波段的遥感数据监测水体表层叶绿素、悬浮物和黄色物质等水色参数^[1]。由于水体的反射率低, 卫星传感器接收到的总辐亮度中能反映水体生物光学特性的离水反射率不足 10%, 而超过 90% 的辐射量来自大气的程辐射, 因此从传感器接收到的总辐射中精确地去除大气影响显得尤为重要, 大气校正成为水色参数反演的重要前提^[2]。

本文基于 ENVISAT 卫星搭载的中分辨率成像光谱仪 (MERIS) 数据, 以太湖为研究区, 展开大气校正的算法研究。利用 6S 辐射传输模型建立一个包含不同几何参数、不同气溶胶光学厚度的模拟数据集; 再根据数据集构建一个卫星辐亮度-离水反射率的神经网络模型, 从而无需同步气溶胶参数即可实现大气校正。

1 数据来源及其处理

1.1 地面实测数据

本文选取的研究区为位于长江三角洲的太湖, 是流域内最大的湖泊, 水域面积为 2 338.1 km², 湖泊面积为 2 427.8 km², 湖岸线全长为 393.2 km^[3]。太湖受陆源物质排放影响严重, 水体特征为浮游藻类和非生物性悬浮物含量较高, 是典型的二类水体^[4]。

本文于 2010 年 8 月 6 日~11 日, 在太湖 66 个测点进行野外测量, 选取的样点均匀分布于太湖中, 如图 1 所示。

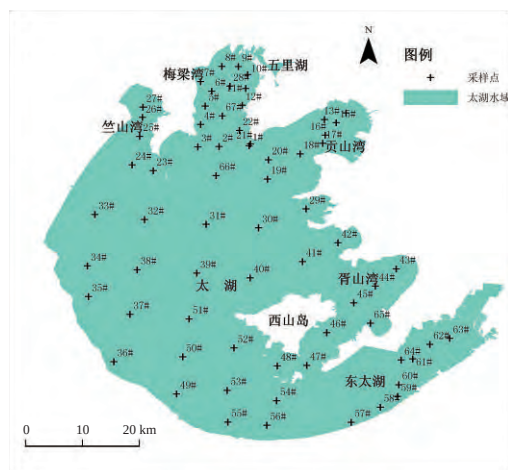


图 1 太湖采样点分布图

1.2 MERIS 影像数据

MERIS 是搭载在 ENVISAT 卫星上的主要传感器之一, 适用于不同尺度的水体研究, 主要检测海岸带复杂二类水以及内陆水质环境。MERIS 从可见光到近红外 (390~1 040 nm) 共设置了 15 个波段, 带宽在 3.75~20 nm 之间。本文选取 2010 年 8 月 9 日太湖境内的 MERIS Level 1p 数据, 先利用 BEAM5.0^[5] 软件对影像进行预处理, 具体流程为: ①消除数据的重叠效应现象; ②以一幅影像为基准图像, 将其他影像与其进行几何配准, 几何校正的同时进行投影转换; ③图像裁剪的目的是去掉研究区外的区域, 按照太湖所在的经纬度范围, 在 BEAM5.0 软件中进行遥感影像的裁剪, 得到太湖的水体全图。

由于 MERIS 的 761 nm 和 900 nm 波段分别位于氧气和水汽的吸收带, 地面的向上辐射信号被严重削弱, 因此这里只研究其他 13 个波段。

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2 大气校正网络的构建

2.1 6S 辐射传输模型

6S 辐射传输模型建立在辐射传输理论的基础上，模拟了太阳辐射在地气系统中的传输过程，采用最新近似和逐次散射算法来计算散射和吸收，以消除大气效应的影响。6S 辐射传输模型广泛应用于大气传输和遥感等领域，受研究区特点和目标类型的影响较小，且适用于市面上大多数卫星遥感产品。

假设地表为均匀的朗伯体，根据 6S 辐射传输模型将卫星辐亮度转化为表观反射率 ρ 的公式为：

$$\rho = \frac{\pi L}{F_0 \cos \theta_s} \quad (1)$$

式中， L 为大气上界辐亮度； F_0 为平均大气上界太阳辐照度； θ_s 为太阳天顶角。

传感器接收到的表观反射率可表示为：

$$\rho(\theta_s, \theta_v, \phi_s, -\phi_v) = T_g(\theta_s, \theta_v) \times [R_{r+a}(\theta_s, \theta_v, \phi_s, -\phi_v) + T(\theta_s)T(\theta_v) \frac{\rho}{1 - RS}] \quad (2)$$

式中， θ_s 为太阳天顶角； θ_v 为太阳方位角； ϕ_s 为卫星传感器观测天顶角； ϕ_v 为卫星传感器观测方位角； $T_g(\theta_s, \theta_v)$ 为波段内大气分子吸收透射率； R_{r+a} 为由瑞利散射和气溶胶散射构成的程辐射； $T(\theta_s)$ 为太阳到地面的下行透过率； $T(\theta_v)$ 为地面到传感器的上行透过率； s 为大气半球反照率； ρ 为地面目标反射率^[6]。

2.2 数据集的构建

本文从准同步的实测离水反射率中随机选取 44 个作为输入的地面数据，建立模拟数据集，剩余的 22 个作为检验数据。神经网络的输入层是卫星接收到的辐亮度，因此训练数据集的辐亮度是将地面数据集中的离水反射率代入 6S 辐射传输模型，再根据不同的几何条件、大气模式、气溶胶模型和气溶胶光学厚度计算得到的。6SV2.0 矢量版本的主要参数包括：

1) 几何观测参数。6SV2.0 能直接从一些常见的遥感图像中获取几何参数，但不包含 MERIS 影像数据，因此必须通过自定义输入太阳和传感器的几何观测参数，包括太阳天顶角、方位角，卫星传感器天顶角、方位角。

2) 大气模式。6SV2.0 提供了热带、中纬度夏季、中纬度冬季、极地夏季、极地冬季、美国标准大气和对流层 7 种大气模式。由于 MERIS 影像成像时间在 8 月，且太湖位于中纬度地区，因此选择中纬度夏季模式^[7]。

3) 气溶胶模型。6SV2.0 提供了大陆型、海洋型和城市型 3 种气溶胶模型。根据 4 种基本粒子成分（沙

尘性、海洋性、水溶性和煤烟性）的混合比，3 种模型所占比例如表 1 所示。

表 1 6S 辐射传输模型中 3 种气溶胶模型基本粒子的比例

Aerosol model	Dust-like	Water-soluble	Oceanic	Soot
Continental	0.7	0.29	0	0.01
Maritime	0	0.05	0.95	0
Urban	0.16	0.61	0	0.22

由于太湖距离东海约 150 km，四周被城市包围，可认为太湖区域几乎不存在海洋性粒子^[8]，因此由表 1 可知太湖气溶胶类型为大陆型或城市型；且吕恒^[9]等国内大部分学者对太湖展开研究时均采用大陆型气溶胶模型，所以本文也采用大陆型气溶胶模型。

4) 550 nm 气溶胶光学厚度。根据对太湖气溶胶光学厚度变化特征的分析，6SV2.0 中气溶胶光学厚度输入区间为 0.3~1.2，步长为 0.1。数据集所需的模拟参数以及它们的阈值范围见表 2。

表 2 MERIS 数据集模拟参数统计

参数	变化范围	步长
太阳天顶角 /°	0~60	10
卫星天顶角 /°	0~60	10
太阳方位角 /°	100~300	20
卫星方位角 /°	100~300	20
大气模型	中纬度夏季	
气溶胶模型	大陆型	
550nm 气溶胶光学厚度	0.3~1.2	0.1
波段	0.412~0.885	

在 6SV2.0 中输入定义的参数，得到 3 个大气校正系数 X_a 、 X_b 和 X_c ，再利用式 (3) 计算校正后的反射率。

$$\rho = \frac{y}{1 + X_c y} \quad (3)$$

$$y = X_a L - X_b$$

式中， ρ 为大气校正后的离水反射率； L 为卫星辐亮度。

2.3 神经网络模型的构建

本文采用的 BP 神经网络模型^[10]，在 Matlab 软件的神经网络工具箱中进行学习计算^[11]。从数据集中随机选取 80% 的数据用于训练，共计 15 840 条；剩余 20% 的数据用于验证，共计 3 960 条，每条数据均包含 13 个波段的卫星辐亮度和离水反射率，构建一个 **卫星辐亮度 - 离水反射率的大气校正神经网络模型**。模型的结构参数如表 3 所示。

表 3 大气校正神经网络参数

模型特征	参数
输入数据	MERIS 辐亮度 B1~B10, B12~B14
输出数据	离水反射率 $\rho_1 \sim \rho_{10}$, $\rho_{12} \sim \rho_{14}$
网络类型	BP 网络
隐含层数	1
隐含层节点数	10
训练函数	LM
传递函数	TANSIG
学习函数	GDM

3 结果分析

本文以 2010 年 8 月 9 日的 MERIS Level 1p 遥感影像为研究对象, 先进行微笑效应和臭氧吸收校正, 再将各波段卫星辐亮度输入该大气校正算法, 进行大气校正, 结果如图 2 所示。

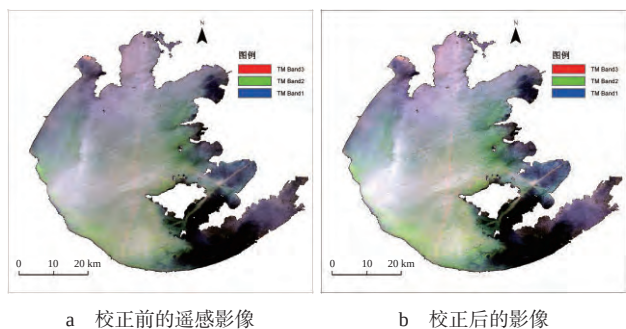


图 2 大气校正前后 MERIS 影像假彩色合成对比图

由图 2 可知, 经大气校正后的影像与原始图像具有相同的特征信息, 基本还原了太湖的真实反射率, 且增加了图像的亮度与清晰度, 比校正前的遥感影像包含了更丰富的信息。

为了分析大气校正前后 MERIS 影像反射率的变化情况, 先利用式 (1) 将采样点的卫星辐亮度转化为表观反射率, 并与准同步的 22 个检测点数据进行对比分析。图 3 为 4 个代表波段大气校正前后反射率的对比情况。经过大气校正后的反射率明显降低, 其中第 2 波段校正后的反射率平均值为校正前的 12%, 第 4 波段为 22%, 第 9 波段为 25%, 第 12 波段为 21%, 说明 MERIS 传感器接收的总辐射信号中, 绝大部分信号来

自于大气散射, 只有一小部分来源于携带水体信息的离水辐射, 因此经过大气校正后的离水反射率在一定程度上去除了大气效应的影响。

为了对本文方法的校正结果进行检验和精度评价, 对 22 个采样点校正后的离水反射率与实测反射率分别求相对误差和标准差, 如表 4 所示。

表 4 MERIS 大气校正精度验证

波段	波长 /nm	相对误差 /%	标准差
1	413	25.03	0.17
2	443	23.69	0.18
3	490	19.11	0.20
4	510	18.13	0.18
5	560	20.75	0.16
6	620	17.83	0.17
7	665	15.35	0.12
8	681	22.36	0.14
9	709	21.00	0.15
10	754	28.50	0.20
12	779	30.34	0.23
13	865	33.79	0.26
14	885	36.27	0.24

由表 4 可知, 所有波段的相对误差中共有 4 个波段低于 20%, 其中第 7 波段最低, 达到了 15.35%, 标准差也是所有波段中最小的, 校正结果具有较好的稳定性; 最后 3 个波段的相对误差高于 30%, 但也没超过 40%。就总体来看, 经本文大气校正算法处理后的 MERIS 影像数据 13 个波段的平均相对误差随波长增加呈先减小后增大的趋势, 其中绿光到红光波段校正精度较好, 相对误差稳定在 10%~20%, 因为该波段水体吸收作用不强, 离水反射率高, 水体信号所占比例

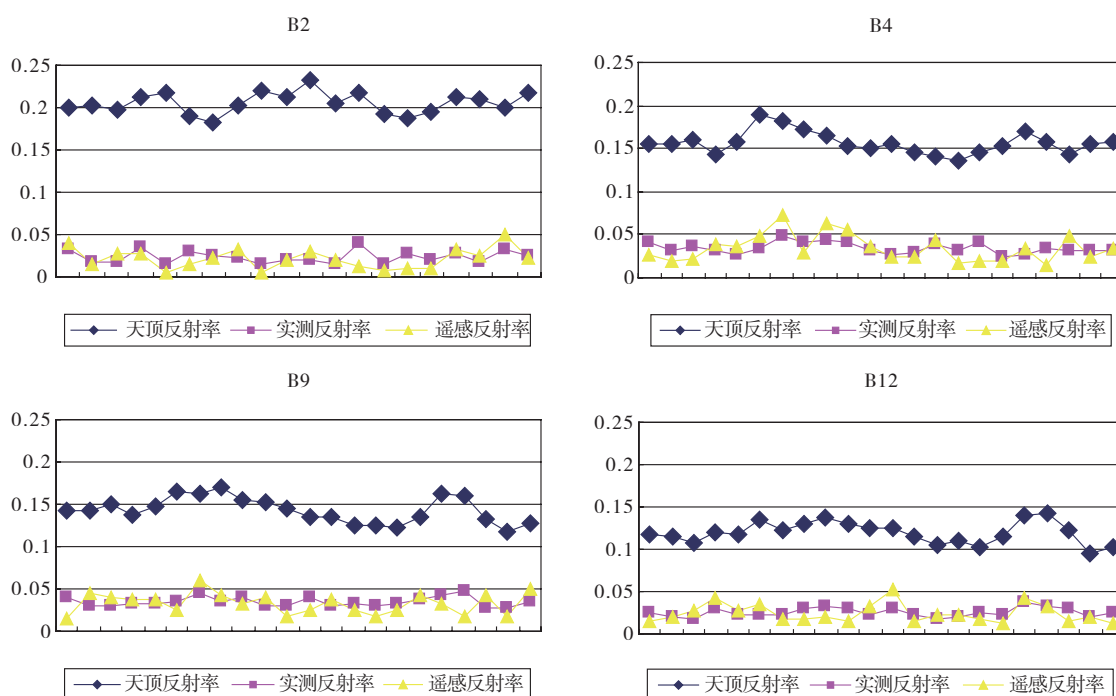


图 3 大气校正前后 MERIS 检验数据集的反射率对比

较高；其次是蓝光波段，相对误差为 20%~30%，该波段精度不高的主要原因为太湖水体中含有大量的叶绿素和黄色物质，在蓝光波段有很强的吸收作用，且大气在蓝光波段有较强的散射作用，导致该波段携带的水体信号几乎淹没在大气散射中；表现最差的是近红外波段，因为该波段水体信号本身的绝对值较低^[12]。

4 结 语

本文以内陆二类水体太湖的 MERIS 高光谱影像为研究对象，针对目前二类水体大气校正中难以获取与遥感影像同步的气溶胶信息的问题，提出将神经网络技术应用于遥感影像大气校正中，成功建立了卫星辐亮度-离水反射率的大气校正模型，无需同步气溶胶参数就能实现 MERIS 数据 13 个波段的大气校正。将校正后的离水反射率与准同步实测的对应采样点进行了对比可知，各波段的平均校正误差分布在 10%~40%，说明本文所述大气校正方法在 MERIS 各波段上的校正精度均较高，稳定性强，对太湖具有一定的适用性。

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(上接第 53 页) 从而实现了管理部门的应急与决策支持。BD/GPS 双模定位可在 5~10 min 收敛到平面 3 mm 和高程 5 mm 的定位精度，能实时或准实时地发现形变位移，对监控中心的进一步模拟形变趋势、及时发现灾害预兆并做好预警和应急方案具有重要作用。

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Building Extraction Method Based on Digital Surface Model

by WANG Ziming

Abstract With assistance of digital surface model (DSM), we proposed a building extraction method by using the combination method of object-based and knowledge-based in this paper. Firstly, we segmented images by multi-resolution segmentation method with the help of DSM to get complete building area. Then, we set three rules to extract buildings, and used the experimental images of ISPRS datasets to verify the method. The results show that the proposed method can effectively extract buildings with a high accuracy, and the commission rate and omission rate are 6.26% and 7.43%.

Key words DSM, building extraction, multi-resolution segmentation (Page:38)

Monitoring and Analysis of Mine Land Occupation in Hunan Province Based on Remote Sensing

by LIU Li

Abstract Based on the remote sensing data of Hunan Province in 2015, we carried out the remote sensing monitoring of mine land occupation in the whole province, and obtained the basic data of mine land occupation. In this paper, based on the characteristics of mining area, mineral resources and mining methods in Hunan Province, we summarized the characteristics of the mine in the province and the main problems of the mine land occupation, and put forward the corresponding countermeasures and suggestions. This study can provide the basis for mine geological environment protection and mining land occupation policy research.

Key words remote sensing, Hunan Province, mine, land occupation (Page:41)

Research on Rapid Construction Method of 3D Virtual Instrument Museum

by DENG Yiguang

Abstract In view of the current situation that virtual instrument museum of 3D digital construction has high cost, long construction period and so on, we put forward a rapid construction method of 3D virtual instrument museum based on general camera. Firstly, we collected a large amount of images with high degree of overlap according to photogrammetry 3D stereovision requirements. And then, we constructed 3D model of instrument museum based on multi-view image dense matching, and realized online publication of 3D virtual instrument museum. Finally, taking the photogrammetric instrument museum of Wuhan University for example, we used this method to construct an online 3D virtual instrument museum, which served the freshmen's professional cognitive teaching, the virtual experiment teaching, the popularization of photogrammetry and remote sensing, and possessed of popularization and application value.

Key words virtual instrument museum, 3D model, rapid construction, virtual experiment teaching (Page:47)

Research on Sensing Monitoring and Network Construction Based on Hybrid BDS/GPS Positioning

by JIANG Changling

Abstract BeiDou satellite navigation system (BDS) is providing services to the Asia-Pacific Region. It will provide services to "One Belt and One Road" and neighboring countries, and gradually to achieve global coverage. BDS plays an important role in many fields, such as surveying and mapping, transportation, monitoring and so on. In this paper, we focused on automatic deformation monitoring in real time using hybrid BD/GPS positioning and Zigbee wireless sensor network. And then, taking a dam for example, we validated the feasibility of this method.

Key words hybrid BD/GPS positioning, sensor network construction, deformation monitoring (Page:51)

Atmospheric Correction for Case-2 Waters Based on Neural Network

by SHEN Juping

Abstract Based on the existing atmospheric-radiative-transfer theory and atmospheric correction methods, we researched the atmospheric correction methods of hyperspectral remote sensing images (MERIS) for inland case-2 water. In this paper, based on the 6S radiative transfer model and neural network technique, we built an atmospheric correction algorithm which extracted water reflectance with satellite radiance, didn't need the synchronous aerosol parameter. And then, we corrected the MERIS data on August 9th, 2010 atmospherically. The comparison between the retrieved reflectance with the measured reflectance showed that the atmospheric effect on the remote sensing imagery was removed, and the water remote sensing reflectance with an average relative error from 10% to 40% in the 13 bands. The retrieved reflectance were close to the measured.

Key words water color remote sensing, MERIS, atmospheric correction, neural network (Page:54)

Design and Realization of Qingdao Tourism Strategy App

by ZHANG Xiang

Abstract Currently, tourism Apps are single in function, and hardly set the route in real time. In view of these, we designed and realized Qingdao tourism strategy App based on the Android platform. The Qingdao tourism strategy App integrates the remote sensing images, tourist feedback information, tourist attractions, hotel catering and other data information, and makes full use of the data information. This App can lead the tourists to the tourist attractions in real time. The tourists can also get the information of the attractions tickets, attractions information and hotel information. So the Qingdao tourism strategy App can help the tourists travel conveniently.

Key words Android, tourism strategy, real time, map (Page:58)

Research on Mathematics Accuracy Detection and Evaluation Technique of the Topographic Map

by QU Huaying

Abstract The traditional topographic map accuracy detection is cumbersome and inefficient. In this paper, we proposed a method of rapid detection and evaluation of topographic map accuracy, based on the demand of rapid detection and process flow. This method processed the accuracy detection in a process-oriented manner. Each detection module consisted of functional components (algorithm), and the final evaluation results were stored by Access database. From the analysis of the example, the conclusion shows that this method can improve the detection efficiency, and is very suitable for the detection and evaluation of the accuracy of digital topographic map.

Key words topographic map, accuracy detection, accuracy evaluation, process flow (Page:61)

Design and Research on API Gateway Based on GIS Service

by LIU Hongbo

Abstract With the speeding up of the national administrative information public step, in order to ensure the security of map service, we put forward API Gateway based on GIS service in this paper. We used Redis, OAuth2 and Nginx to design API Gateway, which included service registry center, service certification center and service agent center. API Gateway provides unified service export, access control and log monitoring for GIS services from different departments, different map service platform.

Key words API Gateway, GIS service, open service (Page:65)

Remote Sensing Monitoring for Human Activity of Provincial Nature Reserve

by LUO Yining

Abstract In this paper, we discussed the monitoring technologies on human activity of provincial nature reserve in Shaanxi Province by RS, which formed the technical route, index, data processing method and key parameters of human activity. On the basis of this, we built the monitoring methods accordingly. Taking Tai'an provincial nature reserve for example, the results show that the image of 5 m resolution should be used first, and then 2 m. There are agricultural land, residential, mining land, quarry, road and other artificial facilities in the nature reserve, which accounts for 17.08% of the total area in 2015. The main human activity type is agricultural land, which area accounts for 86.49% of the total area of human activity. Most of the human activities distribute in experimental area, but there are quarries located in the core area. The area of forest land reduces, and most of the reduced area changed into agricultural land, road and residential from 2013 to 2015. The human activity index of the reserve is 0.001 217 in 2015, and the core area is greatly influenced by human activities, which is not conducive to the ecosystem protection of the reserve.

Key words provincial nature reserve, human activity, remote sensing (Page:68)

Remote Sensing Monitoring of Vegetation Change and Assessment of Damage Degrees in the Opencast Coal Mine Area

by LI Linfeng

Abstract Taking Juhugeng opencast coal mine area in Qinghai Province as the research area, we extracted vegetation coverage by making use of TM images from 2001 to 2016 in this paper. First of all, we analyzed the change trend of vegetation coverage by linear regression of the 6-level data at a time node of every 3 years, and demonstrated the transferring situation between different grades vegetation through the vegetation grade transfer matrix. And then, combining with t distribution, we quantitative assessed the vegetation damage of the study area. At last, we used profile analysis to reveal the situation of restore. The results indicate that the growth of vegetation in the study area shows a negative trend. The constructions of open pit, mine dump and other fields have directly destroyed the original vegetation. High vegetation coverage area has been most severely damaged. The vegetations in the mine surrounding area are negatively correlated with the distance. The restore and governance work has been implemented in the study area, and especially has an obvious effect on the slag mountain area.

Key words vegetation coverage, opencast coal mine, t distribution, damage assessment, restore and governance (Page:72)

3D Velocity Field Analysis of Hefei Based on CORS Data

by REN Jiafeng

Abstract In this paper, we used GAMIT/GLOBK software to process GPS data from HFCORS from January 2011 to December 2016, and acquired the velocity field of the reference stations. Then, based on these results, we analyzed the crustal movement of Hefei City in detail. The results show that the mean horizontal velocity of the HFCORS stations is 34.96 mm/yr, in the direction of E18.69°S under ITRF2014 reference frame. The mean horizontal velocity regard to the Eurasia Plate is 6.06 mm/yr, in the direction of E14.61°S. As for the vertical velocity, the mean vertical velocity is 1.31 mm/yr, and indicates an overall uplift for this area.

Key words reference station, velocity field, ITRF (Page:77)

Design and Realization of Environmental Protection Inspection Information System Based on GIS

by LI Jiong

Abstract In this paper, we analyzed the problems faced by the environmental protection inspection work at first. In order to solve the problems at this stage of the work, we designed different modules according to the user categories, and integrated the existing professional data to standardize the inspection content,